CHEMISTRY

SECTION – I

Single Correct Choice Type

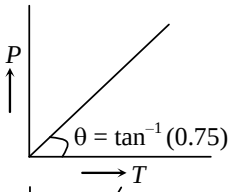
This section contains 8 single correct choice type questions. Each question has 4 choices: (a), (b), (c) and (d) for its answer, out of which **ONLY ONE** is correct.

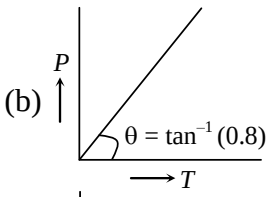
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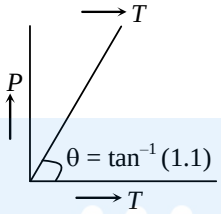
1. 1 mol of gas A dissociates in a closed rigid container of volume 0.16 lit as per the following reaction:

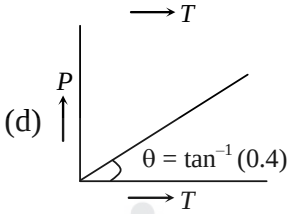
$$2A(g) \longrightarrow 3B(g) + 2C(g)$$

If α for A is 0.4 irrespective of temperature, then the correct P(atm) vs T(K) graph is: [Use R = 0.08 lit at m/mol k)

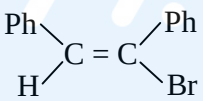
(a) 

(b) 

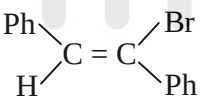
(c) 

(d) 

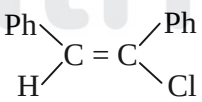
2. Arrange the following compounds according to decreasing rate of elimination with NaOEt/EtOH.



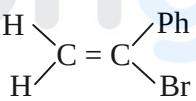
(A)



(B)



(C)



(D)

(a) (A) > (B) > (C) > (D) (b) (B) > (A) > (C) > (D)
(c) (D) > (A) > (C) > (B) (d) (B) > (D) > (A) > (C)

3. 4 g of an unknown monoprotic weak acid is dissolved in water. This solution required 50 mL of 0.4 M NaOH for complete neutralization. When the same solution of monoprotic acid is titrated with 25 mL of given NaOH solution. The pH of resulting solution is found to be 4. The pK_a of the weak acid is

(a) 2.5 (b) 3.0 (c) 3.5 (d) 4.0

4. $(X) + C + Cl_2 \xrightarrow{1000\,K} (Y) + CO$

$(Y) + 2H_2O \rightarrow Be(OH)_2 + 2HCl$

Compound (Y) exist in polymeric chain structure and it is an electron deficient molecule. The compound (Y) is

(a) BeO (b) $BeO.Be(OH)_2$

(c) $Be(OH)_2$ (d) $BeCl_2$

5. The conductivity of a saturated solution of CaF_2 at 18°C was found to be 4.2×10^{-5} S cm^{-1} and the conductivity of water used for making the solution was 2×10^{-6} S cm^{-1}. The molar ionic conductivities at infinite dilution of Ca^{2+} and F^- are 104 and 48 S cm^2 mol^{-1}, respectively. The solubility product of CaF_2 will be

(a) 9.84×10^{-12} M^3 (b) 3.2×10^{-11} M^3

(c) 3.63×10^{-11} M^3 (d) 3.7×10^{-11} M^3

6. Identify the correct sequence of increasing number of π-bonds in the structures of the following molecules:

(I) $H_2S_2O_6$ (II) H_2SO_3 (III) $H_2S_2O_5$

(a) (I), (II) and (III) (b) (II), (III), (I)

(c) (II), (I) and (III) (d) (I), (III) and (II)

7. The Henry's law constant for the solubility of N_2 gas in water at 298 K is 1×10^{-5} atm^{-1}. The percentage of N_2 in air is 80%. The number of moles of N_2 from air that dissolved in 10 moles of water at 298 K and 5 atm pressure is

(a) 4×10^{-5} (b) 4×10^{-4}

(c) 5×10^{-4} (d) 5×10^{-5}

8. A sample containing H_3PO_4 and P_4O_{10} is labelled in the analogous manner as oleum. The mass of P_4O_{10} in a sample labelled as 127%, would be (atomic masses: O = 16, P = 31)

(a) 71 g (b) 27 g

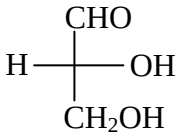
(c) 83 g (d) 35 g

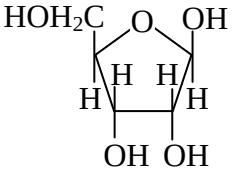
SECTION – II
Multiple Correct Choice Type

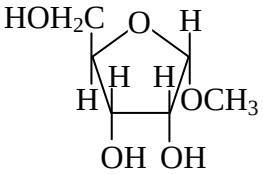
This section contains 4 multiple correct choice type questions. Each question has 4 choices: (a), (b), (c) and (d) for its answer, out of which **ONE OR MORE** is/ are correct.

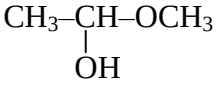
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9. Which of the following undergo aqueous Br_2 oxidation?

(a) 

(b) 

(c) 

(d) 

10. Identify the incorrect statement(s)

(a) Solvay process cannot be used to prepare K_2CO_3 due to precipitation of $KHCO_3$

(b) The gas evolved during commercial preparation of NaOH using Castner-Kellner cell is oxygen and H_2

(c) All alkaline earth metal halides are ionic in nature

(d) The hemihydrate of calcium sulphate is known as gypsum

11. The equilibrium between three gaseous isomers A, B and C can be represented as:

$$A(g) \rightleftharpoons B(g) \quad K_1 = ?$$
$$B(g) \rightleftharpoons C(g) \quad K_2 = 0.4$$
$$C(g) \rightleftharpoons A(g) \quad K_3 = 0.6$$

If initially one mole of A is taken in a closed vessel of volume 1 litre, then which of the following is/are correct statements?

(a) [A] + [B] + [C] = 1 M at any time during the reaction.

(b) Concentration of [C] is $\frac{1}{2.1}$ M at equilibrium.

(c) $K_1 = \frac{1}{0.24}$

(d) Isomer (A) is least stable thermodynamically.

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12. Which of the following is/are correct about $C_3H_4Cl_2$ (A) ?

(a) (A) has total number of isomers as 11.

(b) All isomers of (A) are reducible and give addition products.

(c) All isomers of (A) can undergo free radical substitution reactions.

(d) All isomers of (A) cannot give electrophilic substitution reactions.

SECTION–III
Comprehension Type

This section contains 2 groups of questions. Each group has 3 multiple choice questions based on a paragraph. Each question has 4 choices: (a), (b), (c) and (d) for its answer, out of which **ONLY ONE** is correct.

Passage-I

Metallic gold is frequently found in aluminosilicate rocks and it is finely dispersed among other minerals. It may be extracted by treating the crushed rock with aerated sodium cyanide solution. During this process, metallic gold is slowly converted to $[Au(CN)_2]^{2-}$, which is water soluble, the metallic gold is recovered from it by reacting the gold complex with zinc, which is converted to $[Zn(CN)_4]^{2-}$.

Gold in nature is frequently alloyed with silver, which is also oxidised by aerated sodium cyanide solution, similar to gold.

500 L of a solution of 0.01 M $[Au(CN)_2]^-$ and 0.003 M $[Ag(CN)_2]^-$ was evaporated to one third of the original volume and was treated with 40 g zinc. Assuming that deviation from standard conditions is unimportant in this case and that all these redox reactions go essentially to completion. Given:

$$[Zn(CN)_4]^{2-} + 2e^- \longrightarrow Zn + 4CN^- \quad ; \quad E° = -1.26\ V$$

$$[Au(CN)_2]^- + e^- \longrightarrow Au + 2CN^- \quad ; \quad E° = -0.60\ V$$

$$[Ag(CN)_2]^- + e^- \longrightarrow Ag + 2CN^- \quad ; \quad E° = -0.31\ V$$

13. The concentrations of $[Ag(CN)_2]^-$ after the addition of Zn in it, would be

(a) 0.03 M (b) 0.003 M

(c) 0.0018 M (d) 0.018 M

14. What concentration of cyanide ion is required to keep 99 mol % of the gold in solution in the form of the cyanide complex? Given: K_f of $[Au(CN)_2]^-$ is 4×10^{28}.

(a) 2×10^{14} M (b) 5×10^{-14} M

(c) 2×10^{-14} M (d) 5×10^{14} M

15. There have been several efforts to develop alternative gold extraction processes, which would replace the present one. The reason for this is

(a) NaCN solution corrodes mining machinery.

(b) gold obtained by this process is impure.

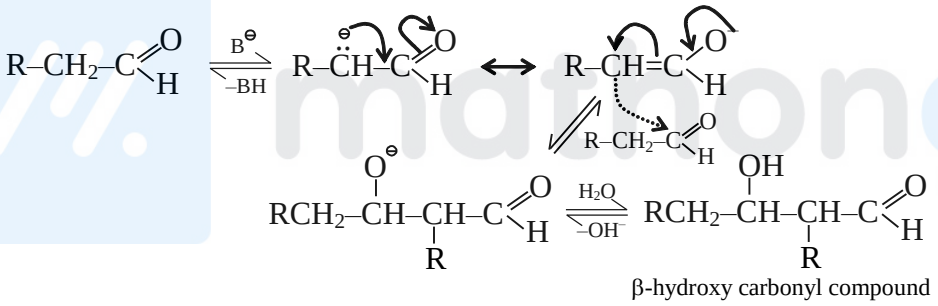
(c) silver is precipitated faster than gold.

(d) NaCN escapes into ground water and releases HCN.

Passage-II

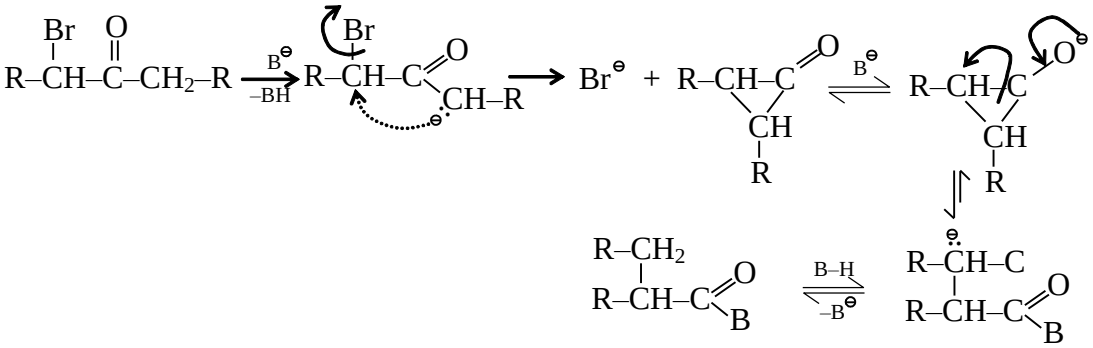
OH^-/OR^- can act as base as well as nucleophile in its reactions. In its reactions with carbonyl compounds (aldehydes, ketones, esters) it act as base and give various condensation and rearrangement reactions.

It undergoes aldol type reaction (OH^- act as base in such reactions) by the following route.

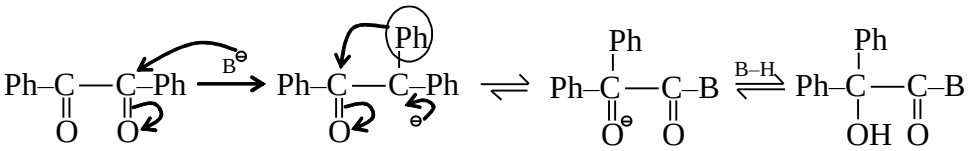


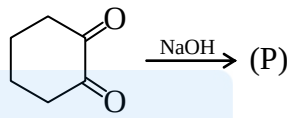
β-hydroxy carbonyl compound

It also undergoes favorski type reactions (OH^-/OR^- act as base in such reaction) by the following pathway.

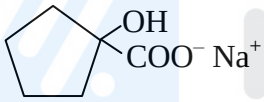


It can also undergo Benzil-Benzilic acid rearrangement (OH^- act as a nucleophilie) as per the given mechanism.

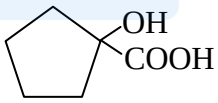


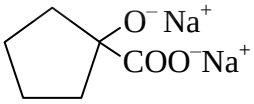
16.  (P)

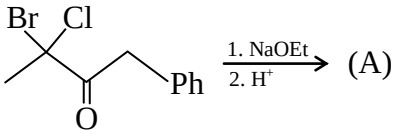
The product (P) is

(a) 

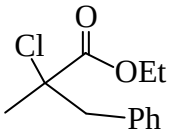
(b) 

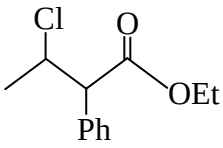
(c) 

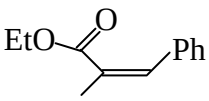
(d) 

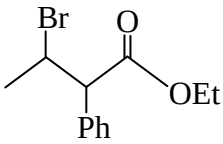
17. 

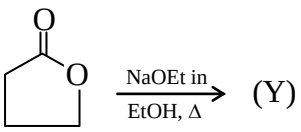
The product (A) in the given reaction would be

(a) 

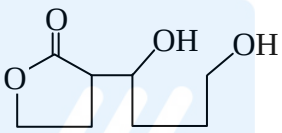
(b) 

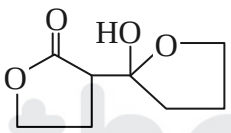
(c) 

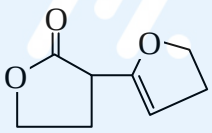
(d) 

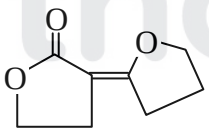
18. 

The product (Y) of the given reaction is

(a) 

(b) 

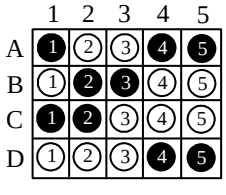
(c) 

(d) 

SECTION–IV
Matrix–Match Type

This section contains 2 questions. Each question contains statements given in two columns, which have to be matched. The statements in **Column I** are labelled A, B, C and D, while the statements in **Column II** are labelled 1, 2, 3, 4 and 5. Any given statement in **Column I** can have correct matching with **ONE OR MORE** statement(s) in **Column II**. The appropriate bubbles corresponding to the answers to these questions have to be darkened as illustrated in the following example:

If the correct matches are A – 1, 4 and 5; B – 2 and 3; C – 1 and 2; and D – 4 and 5; then the correct darkening of bubbles will look like the following.

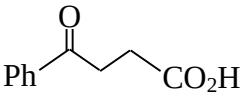
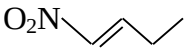
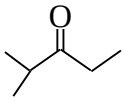


1.

Column I	Column II
(A) $Sb^{3+} \rightarrow (SbO)^{+}$	**1.** Acidification
(B) $CrO_4^{2-} \rightarrow Cr_2O_7^{2-}$	**2.** Addition of base
(C) $MnO_4^{-} \rightarrow MnO_4^{2-}$	**3.** Hydrolysis
(D) $B_4O_7^{2-} \rightarrow B(OH)_3$	**4.** Colour change
	5. Reduction

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2.

Column I (Compounds)	Column II (Oxidising/reducing agents)
(A) R–CH=N–R	**1.** SeO_2
(B) 	**2.** Zn–Hg/HCl
(C) 	**3.** $NaBH_4$
(D) 	**4.** H_2–$[RhCl(Ph_3P)_3]$
	5. CF_3CO_3H

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